

Carlos Molina Martínez

Data Engineer in training · Industrial Engineering & AI

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PROFILE

Engineering student completing a **Double Master's in Industrial Engineering & Artificial Intelligence**, transitioning into **Data Engineering and Industrial Analytics**. The Industrial Engineering master grants the **professional qualification as an Industrial Engineer** on top of a BSc in Energy Engineering. I combine real domain knowledge of industry and energy systems with hands-on skills in **Python, data preprocessing, Machine Learning, LLMs/RAG and MLOps**. I am a proactive, fast-learning engineer, genuinely eager to take on **any opportunity** — graduate program, internship or entry-level role — where physical infrastructure produces large-scale data.

EDUCATION

Double Master's in Industrial Engineering & Artificial Intelligence

2025 – 2026 (EXPECTED)

Universidad Loyola — Seville, Spain

- The Industrial Engineering master is the **professionally accrediting degree (Ingeniero Industrial)**: industrial constructions & installations, automation of systems, manufacturing systems, instrumentation & electronics, thermal & hydraulic installations, structural analysis, chemical process industry and management of organisations.
- AI master: **data preprocessing & wrangling**, machine learning and **deep learning, LLMs and RAG**, computer vision (OpenCV, YOLO) and **MLOps** — experiment tracking with MLflow, **CI/CD pipelines** and cloud deployment.

BSc in Energy Engineering — Specialisation in Renewable Energies

2020 – 2025

Universidad de Sevilla & Universidad de Málaga — Spain

- Solid grounding in **energy technology and industry**: power systems and electrical engineering, thermodynamics and heat transfer, fluid mechanics, and renewable energy technologies (solar, wind, hydro) with their integration into the electrical grid.
- Erasmus exchange year** at Alma Mater Studiorum — Università di Bologna, Italy (2023–2024).

EXPERIENCE

Engineering Intern — WATS Técnicas de Ingeniería

OCT 2024 – APR 2025

Seville, Spain — renewable energy (solar PV) and civil works

- Drafted technical projects for **low-voltage electrical installations** and **photovoltaic generation plants** — engineering calculations, schematics and regulatory documentation.
- Automated day-to-day project calculations with AI/Python**, cutting repetitive work and turnaround time.
- On-site visits as part of the **project supervision team (dirección facultativa)**; technical reporting and project management under strict deadlines.

Student Council & Activities Coordinator — C.M.U. San Juan Bosco

OCT 2022 – JUN 2023

Seville, Spain

- Led a team of **23 delegates** organising educational and cultural activities; president of the Entrepreneurship Club for two years.

PROJECTS & AWARDS

1st Place — AI Challenge, 4th Ogathon (OGA.ai)

JAN 2026

- Won the AI category: predictive model on **60,000 real clinical records** with a heavily imbalanced target. Noise cleaning and a tuned **XGBoost** model, built in under 3 hours.

Railway Fault Diagnosis with NLP — X Hackathon for Smart Industry

Nov 2025

Universidad Pontificia Comillas ICAI × CAF × Kearney — representing Universidad Loyola

- 24-hour challenge: **embedding-based NLP system** to interpret noisy, unstructured maintenance communications between technicians and operators — robust to outliers and usable by field staff.

TECHNICAL SKILLS

Programming & Data	Python (advanced, AI/Data focus), Pandas, NumPy, data wrangling & preprocessing, Docker, Prefect, Git, SQL
Machine Learning & AI	Scikit-learn, PyTorch, XGBoost, deep learning, NLP / embeddings, LLMs & RAG
Computer Vision	OpenCV, YOLO, CNNs, image pipelines
MLOps & CI/CD	MLflow, CI/CD pipelines, GitLab automations, cloud deployment (AWS)
Visualization & Reporting	Power BI (advanced), DAX, Matplotlib, Seaborn, MS Office (Word, Excel, PowerPoint)
Engineering Software	AutoCAD, Civil3D, RFEM, MS Project, PVGIS, pvlb

LANGUAGES & ADDITIONAL INFORMATION

Languages	Spanish (native) · English (B2, preparing C1; lived in the USA and Ireland) · Italian (basic, one academic year in Italy)
Leadership Training	“Sputnik 2021”, “Curious Minds” (Fundación Valentín de Madariaga), HIL Program (Universidad de Sevilla)
Mobility	Open to relocation (EU & international) · Driving licence (B)
Interests	IoT, generative AI, cloud computing, sustainability, hiking, rugby, guitar

References available upon request · Portfolio: carlosmolina55.github.io